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Foundations of NLP for Real-World Data: Tasks, Models, and Research Applications

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Opioid overdose: an important public health challenge

Background

- Opioid-related mortality in Australia has doubled in the past decade.
- Policy responses have been introduced to address growing opioid-related harms.
- Opioid overdose require urgent treatment, therefore, emergency department (ED) data offer valuable surveillance opportunities.

The problem

- Presentations may involve pharmaceutical opioids, heroin or multiple substances.
- The recorded diagnosis may not clearly identify the substance involved.
- Inadequate structured ED data limits opioid harm surveillance and evaluation of the intended and unintended effects of policy interventions.

The opportunity

- Can NLP help us identify opioid overdose presentations more accurately using routine ED records?

What do structured health records tell us?

Person ID	Presentation date	Arrival time	Age	Sex	Postcode	Triage category	Principal diagnosis (ICD-10)	Departure status	Length of stay (hrs)
000123	2025-06-02	14:35	41	F	3000	3	S52.5 (fracture of lower end of radius)	Treated and discharged	3.8
000456	2025-06-05	02:10	53	M	3141	2	S06.0 (concussion)	Admitted	5.5
000789	2025-06-05	19:50	3	M	3053	3	T39.1 (poisoning by 4-aminophenol derivatives, e.g. paracetamol)	Treated and discharged	4.5
000123	2025-06-19	09:15	41	F	3000	4	S61.0 (open wound of finger, without damage to nail)	Treated and discharged	2.1

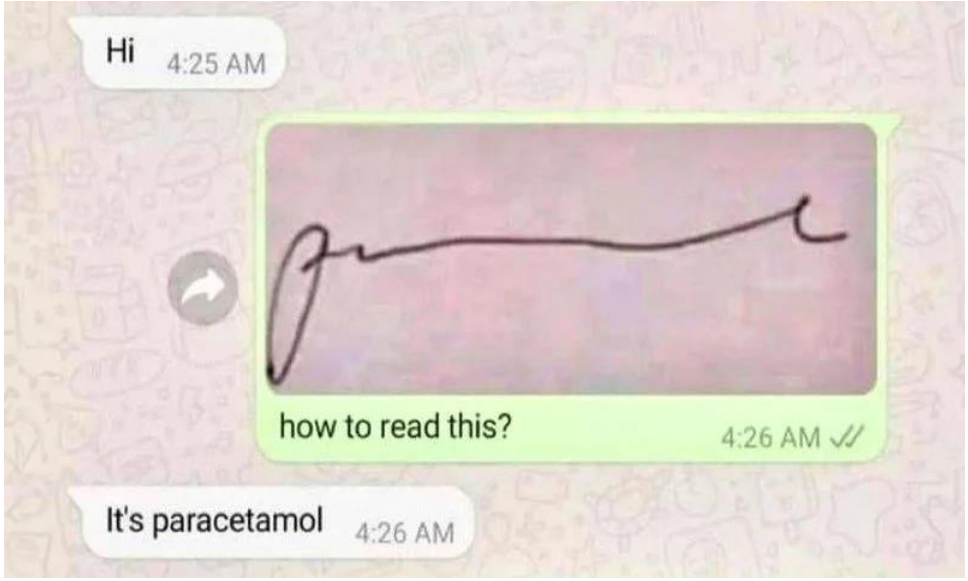
Example notes are fictional

What can unstructured clinical notes tell us?

Person ID	Presentation date	Event description
000123	2025-06-02	fell off bike onto outstretched hand approx 1hr ago. Wrist very swollen and deformed, unable to move fingers without pain. No head strike, no LOC. Pt anxious, asking for pain relief. Pain 7/10. Obs stable. Sent to Xray.
000456	2025-06-05	male brought in by ambulance after being hit in the head during a fight outside a pub. Brief LOC per friend. Vomited x2 on arrival, drowsy but rousable. GCS 14. For CT brain.
000789	2025-06-05	Mum states 3yo found with open bottle of Panadol Elixir in the pantry, unsure how much was taken, maybe half the bottle. Child alert, no vomiting. Poisons Info contacted, bloods ordered.
000123	2025-06-19	back again, cut finger on kitchen knife while cooking. Bleeding controlled with pressure, wound approx 2cm, tetanus up to date.

Example notes are fictional

The challenge of real-world text



Baba- delib od @ 1820 of tepentadol (? X6 ir unknown dose), lorazepam 5x0.5, olanzapine 10x10mg

?opioid OD, trial of naloxone, positive response

'G' at club, found unrousable, vomited, GCS 7

Example notes are fictional

Why unstructured information matters too

Richer detail

Notes record how the injury happened, what was taken and how much.

"fell off bike onto outstretched hand "

Fills gaps in the codes

When the drug or intent is unclear, the code is vague. The note often holds clues.

"tepentadol (? X6 ir unknown dose), lorazepam 5x0.5, olanzapine 10x10mg"

Context and clues

Treatments and responses hint at what happened, such as naloxone for a suspected opioid overdose.

"trial of naloxone, positive response"

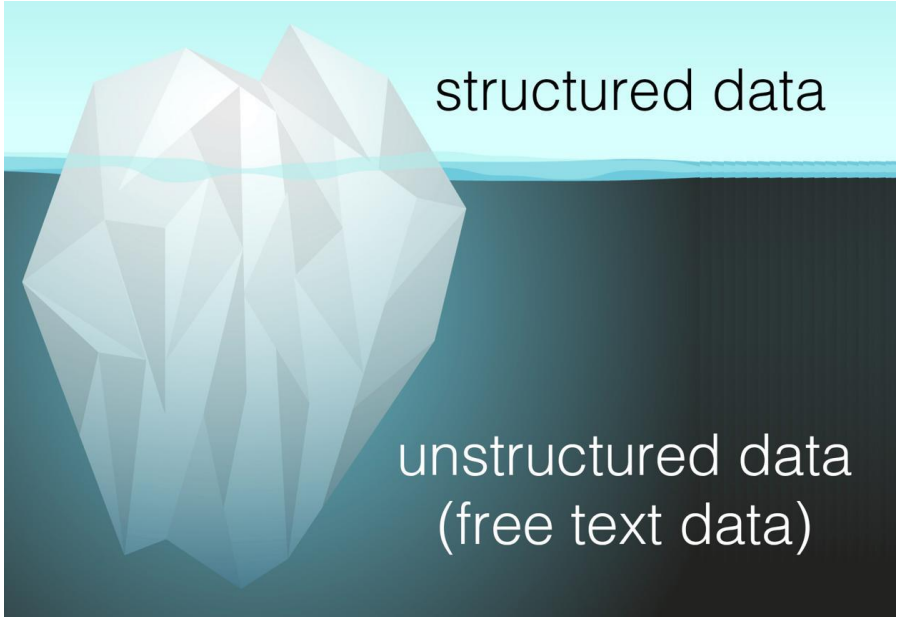
Spots new patterns

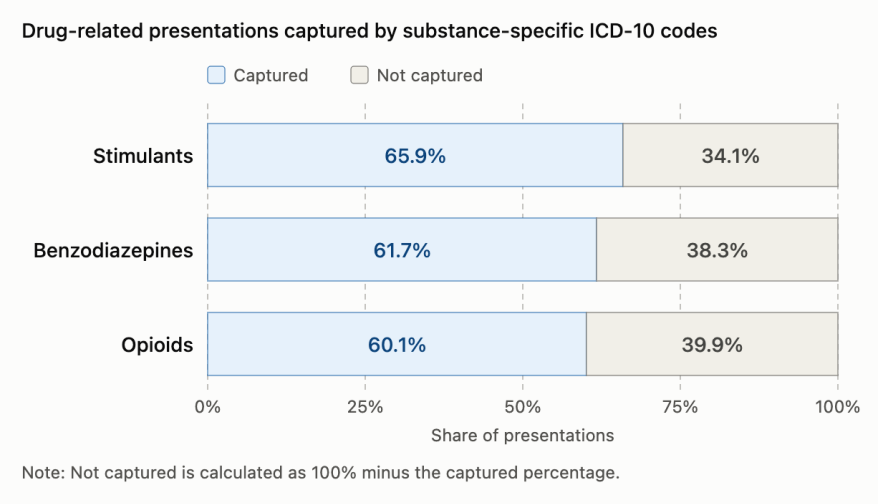
New substances and street names can appear in free text before a code exists.

"'G' at club, found unrousable, vomited, GCS 7"

Structured data counts what happened. Free text often explains how and why.

From clinical text to meaningful data





Example note: ?opioid OD, trial of naloxone, positive response



How a human annotator determines case status?

1. Does the note describe overdose or poisoning?

↓

No
Assess whether the text provides interpretable clinical evidence

Yes
Identify the substance involved

↓

2. Is opioid involvement documented?

✓
Yes
Opioid poisoning

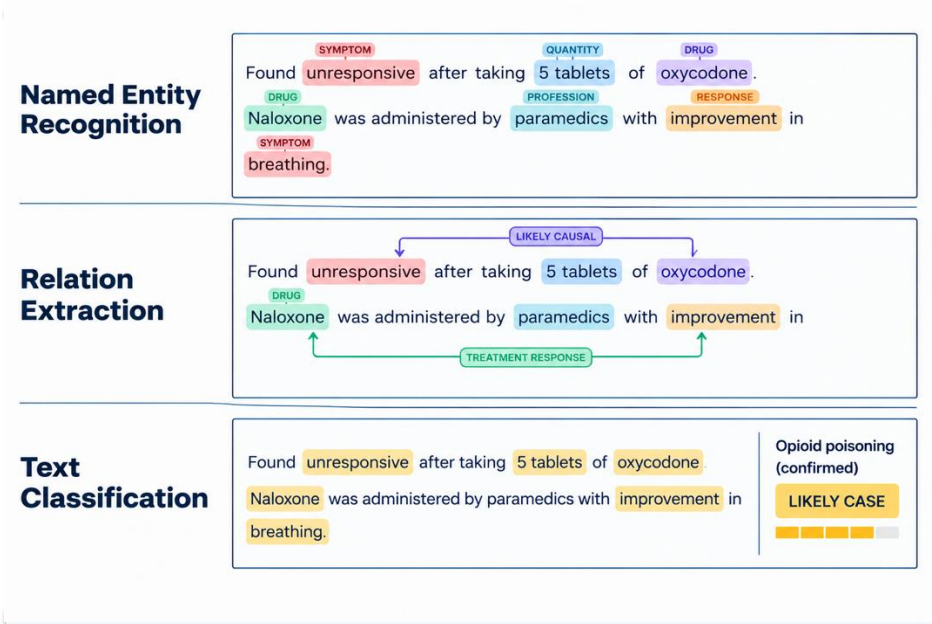
?
Uncertain
Possible opioid poisoning

×
No
Other drug or no opioid poisoning

How NLP supports the same process?

Human reviewer	NLP system
Find relevant details	Extract entities
Recognise clinical features	Detect concepts
Interpret “no overdose”	Detect negation
Establish timing	Detect temporal context
Integrate multiple clues	Combine features
Apply case definition	Classify the case

What does Natural Language Processing (NLP) actually do?



Extract	Identify drugs, symptoms and treatments
Classify	Label a sentence or a whole note, such as a poisoning case
Relate	Link a drug to an event, symptom or response
Interpret	Capture negation, uncertainty and timing

These are complementary tasks—not necessarily four sequential steps in every system.

How much agreement is enough?

From human annotation to NLP validation

Human ↔ Human

Inter-rater reliability

Can two researchers apply the same case definition consistently?

Cohen's kappa · Percent agreement

Human ↔ NLP

Model performance

How well does the NLP model reproduce the validated labels?

Sensitivity · Specificity · PPV · F1

NLP and ML applications in psychoactive drug overdose research - scoping review

2,111 records from four databases → **1,922** screened after removing 189 duplicates → **42** full texts assessed, 31 excluded → **11** studies included

11/11	studies conducted in the US, published between 2019 and 2026
11/11	focused primarily on opioids; only 3 also examined other substances such as benzodiazepines and stimulants
6/11	analysed clinical notes from EHRs; the rest used death records (3), EMS records (1) or a national hospital survey (1)
8/11	used NLP + ML
9/11	used manual chart review or expert annotation as the reference standard.
9/11	reported PPV/precision.

Scoping review

Methods used

1. **Rule-based NLP and NER tools**
   cTAKES, spaCy, MedCAT, QuickUMLS
2. **Classical machine learning**
   SVM, logistic regression, XGBoost (4 studies each)
3. **Embeddings and neural networks**
   TF-IDF, GloVe, CUI2Vec, BiLSTM-CRF
4. **Transformers and LLMs**
   BERT, BioClinicalBERT, GatorTronGPT, Llama 3

Best result reported by selected studies

Study and task	Best model	Result
Ward 2023 Substances in overdose deaths	BiLSTM-CRF with Flair embeddings	**F1 0.99**
Funnell 2026 Substances in overdose deaths	Fine-tuned BioClinicalBERT	**F1 0.97 (external data)**
Lenert 2022 Opioid-related sentences	BERT plus rules	**F1 0.94**
Graham 2024 Opioid overdose	Custom flags + Ensemble	**AUROC 0.92**
Sharp 2025 Predicting opioid overdose death	QuickUMLS+XGBoost,	**F1 0.65**

Tasks, datasets and reference standards differ, so results are not directly comparable.

Where performance was weaker? Less common substances; missing or limited notes; indirect clues

NLP and ML applications in psychoactive drug overdose research – ED presentations involving opioid

Aims

To develop and internally validate a pipeline that applies NLP and ML classifiers to identify opioid overdoses in routinely collected ED data

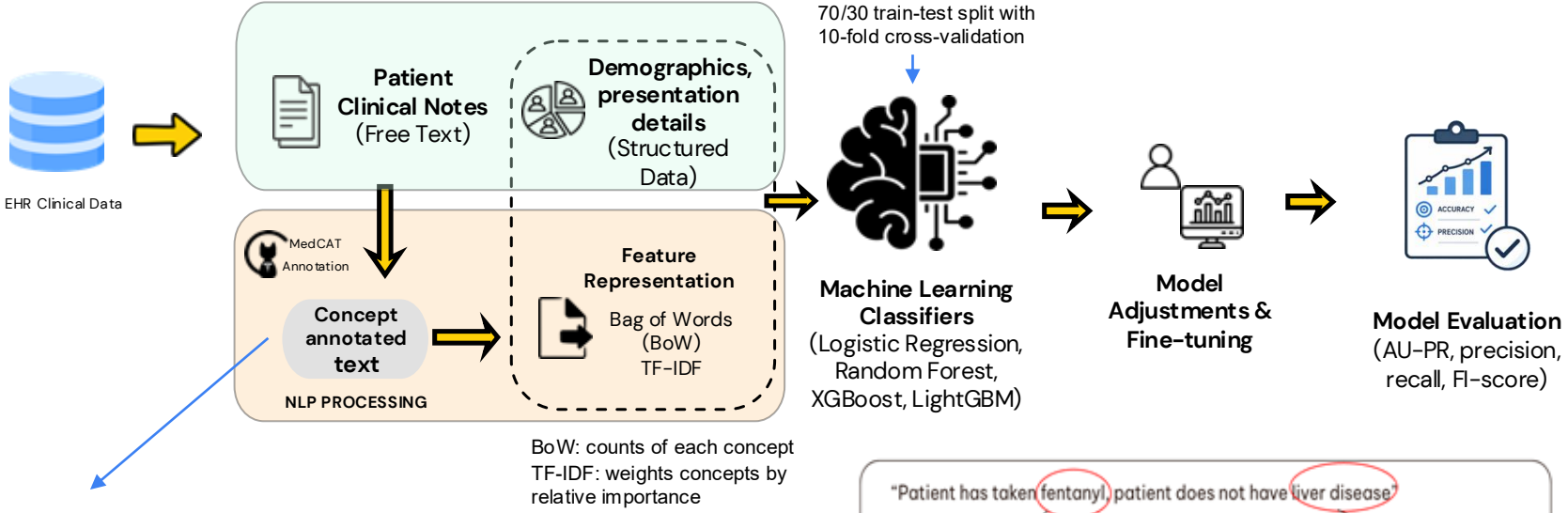
Data source

Victorian Emergency Minimum Dataset (VEMD; VISU): De-identified injury-related ED presentations in Victoria (July 2009-June 2024), including patient demographics, presentation details, diagnoses, outcomes, and a 250-character free-text injury description.

Methods

1. Ground truth development based on free-text descriptions, cause of injury and diagnosis fields.
2. Two-stage model development: NLP feature extraction followed by ML classification.

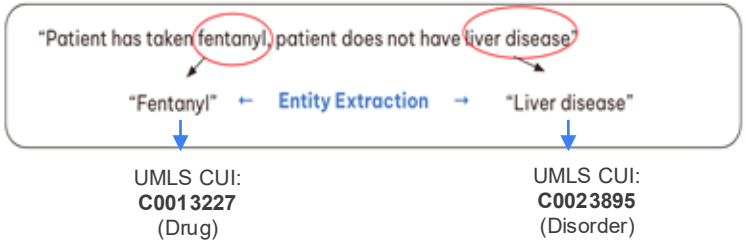
Model Development



BoW: counts of each concept
TF-IDF: weights concepts by relative importance

<u>What is Concept Annotation?</u>

Identifying concepts by understanding the context surrounding in unstructured text fields, and links to biomedical ontologies such as SNOMED-CT or Unified Medical Language System (UMLS).



Results

5.7M Total ED Records

15,678 Opioid Overdose Cases

328,613
Unique Text Words

10 Words
Avg. Text Length

59 Chars
Avg. Character Length

Results

	Accuracy	Precision	Recall	F1-score
	Overall % of correct cases predicted by model	% of actual positives in all model positives	% of correct model predictions in all actual positives	Mean of precision & recall
ICD-10-AM Diagnosis	**0.99**	0.75	0.12	0.20
Model 1: Structured only	**0.89**	0.85	0.02	0.04
Model 2: Structured + Text	**1.00**	0.68	0.88	0.77

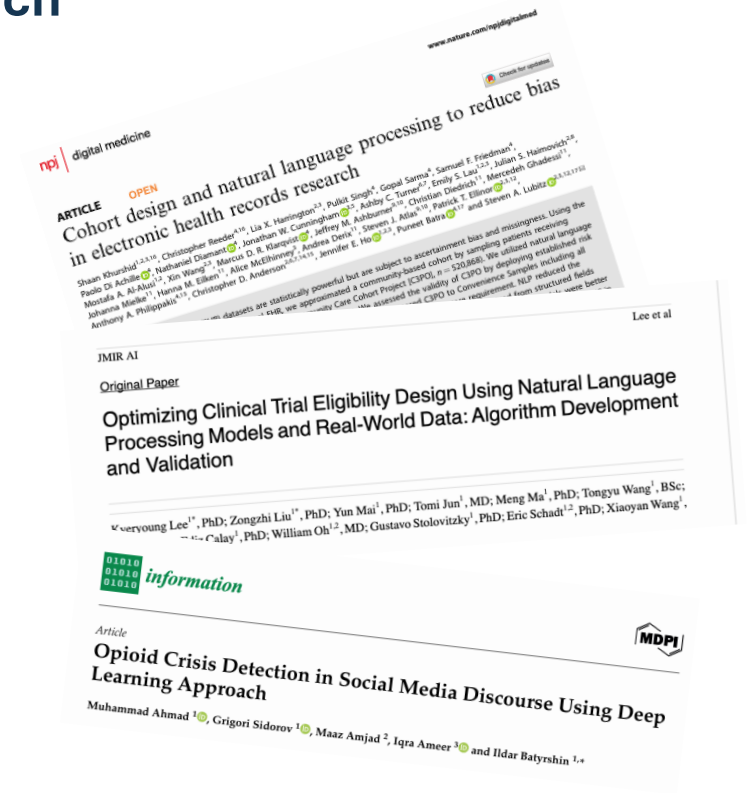
What are the challenges?



- Unknown or unspecified substance ingestion
- Symptoms suggestive of poisoning but no confirmed drug
- Negation, historical use and uncertain timing
- Incomplete documentation and differences between hospitals

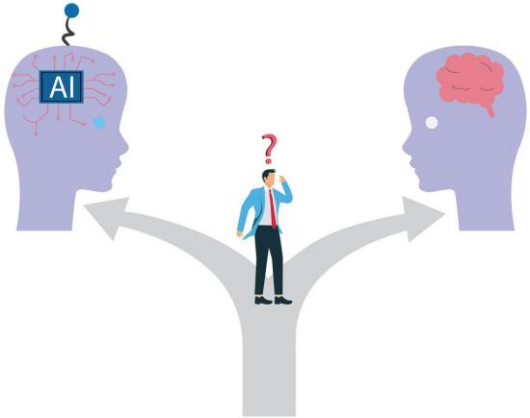
Beyond overdose: NLP across health research

- Cohort identification: Find eligible patients using clinical features that are not captured in codes.
- Clinical trial recruitment: Identify potentially eligible participants from routine records.
- Social media and public health surveillance: Detect emerging concerns, experiences and signals.



Choosing the right approach: rules, ML or LLMs?

- Can you write the criteria down?
- Do you have labelled examples?
- Is the language varied or context-dependent, and do governance and compute allow it?
- In every case, how will you validate it?



- Model selection depends on the research question, available labels, data governance, computational resources and validation requirements.
- More complex models are not automatically better.